JACOB HOGUE

Geospatial Engineer • Full-Stack Data & Cloud Systems • Python & SQL • Customer-Facing Technical Solutions
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PROFESSIONAL SUMMARY

Geospatial Engineer with 8+ years of experience building production data systems, full-stack web applications, and customer-facing technical solutions on Azure. Currently working through a structured AI/ML engineering roadmap covering deep learning, RAG, multi-agent systems, and production MLOps. Track record delivering cloud-deployed platforms with CI/CD automation, managing enterprise databases (PostgreSQL, SQL Server), and teaching technical concepts to audiences ranging from municipal government clients to graduate students. Combines strong software engineering fundamentals with customer-facing experience and a growing expertise in AI and machine learning.

CORE COMPETENCIES

Professional (Production)

• Python & SQL	• CI/CD (GitHub Actions)	• Customer-Facing Consulting
• JavaScript & REST APIs	• Docker	• Technical Training & Mentoring
• Azure Cloud Infrastructure	• Flask & Full-Stack Web Apps	• Geospatial Analysis & GIS
• PostgreSQL / PostGIS	• Data Pipeline Engineering	• Production System Deployment
• Deep Learning (ESRI DL Packages)	• LiDAR & Image Segmentation	• 3D Visualization (Cesium)

Tools: Azure, Google Cloud Platform (GCP), Firebase, PostgreSQL, PostGIS, Microsoft SQL Server, Git, Docker, Flask, REST APIs, GitHub Actions, CI/CD (Continuous Integration/Continuous Deployment), TypeScript, R, ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, GDAL, Geopandas, H3, Parquet, GeoJSON, Cesium, QGIS, Supabase, ETL, Data Pipelines, Valhalla, routingpy, ESRI Network Analyst, OSMnx

AI / ML (Learning)

• NumPy & Linear Algebra	• Anomaly Detection	• SVD / Matrix Decomposition
• Time Series Analysis	• Data Pipeline Design	• Async Data Ingestion

Tools: scikit-learn (sklearn), NumPy, Pandas, Matplotlib, HuggingFace, Jupyter, Conda, Claude API (Anthropic), GPT (OpenAI), Machine Learning (ML), Artificial Intelligence (AI), Windsurf, HeyGen, BeautifulSoup, PostMan

PROFESSIONAL EXPERIENCE

Airspace Link | Remote

Geospatial Engineer September 2022 – Present

Full-stack engineer on a cloud-deployed geospatial platform serving UAS stakeholders across municipal, enterprise, and federal clients.

- Built and maintained production geospatial platform on Azure, writing the data pipelines and REST API services that serve spatial features to enterprise and government stakeholders
- Led spatial risk assessment for City of Romulus, MI that elevated 34 FAA ceiling grids and expanded drone flight area by 13.92 square miles, improving UAS operational access for 5,500+ residents and ~300 businesses
- Designed automated workflow (Survey123 + Power Automate) generating 25-page tailored community drone readiness reports, reducing report generation time by ~50% and lowering repeated sales engagement cycles
- Applied H3 indexing, routing analysis, and airspace modeling for City of Arlington, TX multimodal delivery study and contributed to HexGen Risk Assessments scoring airspace risk across municipal and federal jurisdictions
- Implemented ESRI deep learning packages to extract building footprints from imagery and perform LiDAR-based analysis of tree cover for airspace obstruction modeling
- Built a drone flight simulator using Cesium for 3D airspace visualization and flight path planning
- Built a Fire Incident Response and EMS Dashboard for the Detroit Aerial Innovation Region project, comparing real personnel driving response times against simulated drone-as-first-responder (DFR) response times from stations to incidents
- Follow and maintain code standards across a full CI/CD deployment pipeline (GitHub Actions, Docker) with consistent deployments across Development, Staging, and Production environments on Azure, ArcGIS Online, and PostgreSQL

City of Rochester, New Hampshire | Rochester, NH

GIS / Asset Management Technician March 2021 – September 2022

Technical lead for GIS infrastructure, data management, and application development serving cross-departmental city operations.

- Built production web applications in Python, SQL, and JavaScript that replaced manual work order and service request workflows for 100+ city staff
- Managed enterprise geodatabases and Microsoft SQL Server, designing optimized data pipelines and SQL view tables for cross-departmental data accessibility
- Built data-driven fleet management application integrating real-time IoT sensor data from Verizon Fleet Management, enabling data-informed budgeting decisions and cost control
- Translated spatial data into clear deliverables for engineering and environmental design projects used by non-technical stakeholders

Granite State Rural Water Association | Walpole, NH

GIS / Source Water Specialist May 2020 – March 2021

Customer-facing technical consultant providing on-site GIS training and data analysis to rural communities across New Hampshire.

- Provided on-site technical consulting and training to rural and small community stakeholders, breaking down geospatial and data concepts into practical source water protection plans they could implement independently
- Coordinated cross-agency collaboration with state, county, and local government partners to identify, prioritize, and implement environmental protection programs
- Produced hydrology, geology, and soil analysis deliverables for the Whitewater Watershed Management Plan, used directly in resource management planning

Antioch University New England | Keene, NH

GIS Research & Teaching Assistant August 2018 – May 2020

- Taught Intro to GIS and Advanced GIS courses to Master's students with backgrounds ranging from complete beginners to experienced professionals
- Conducted spatial analysis research using Python and advanced analytical tools at the Antioch Spatial Analysis Lab, delivering projects for ecological consulting firms and environmental nonprofits
- Developed data collection applications for field research organizations including the American Turtle Observatory, enabling real-time species tracking and data capture

New Hampshire Geological Survey | Concord, NH

Stream Crossing Assessor May 2019 – August 2019

FEMA Hazard Mitigation-funded statewide stream crossing assessment program under NH DES Hazards Program.

- Collected and assessed stream crossing data statewide using geomorphic compatibility indices, feeding hydraulic/hydrologic models for infrastructure risk evaluation
- Processed SSURGO and LiDAR DEM data using ArcHydro tools for watershed delineation and input into hydraulic assessment models
- Performed QA/QC of data in the Statewide Asset Data Exchange System (SADES), ensuring data integrity across the statewide assessment program
- Trained Regional Planning Commission staff on the Statewide Stream Crossing Assessment Protocol

Fitchburg State University | Fitchburg, MA

Three undergraduate research positions across planetary science, public safety analytics, and environmental hydrology.

Research Intern (Dr. Reid Parsons) June 2016 – August 2016

- Mapped extent and ice-flow characteristics of Martian lobate debris aprons as part of a joint grant with NASA Ames Research Center
- Conducted solar irradiance assessment of Fitchburg State campus buildings using ESRI solar irradiance tools

Research Intern (Dr. Jane Huang) September 2016 – May 2017

- Performed hotspot and cluster analysis mapping of crimes, foreclosures, and building permits for the ReImagine North of Main Community Revitalization Project, collaborating with Crocker Center and Fitchburg Police Department
- Presented findings at NEURISA Day 2016 and Fitchburg State Undergraduate Conference on Research and Creative Practice (2017)

Senior Research Assistant, GIS (Dr. Jane Huang) May 2017 – August 2017

- Conducted Nashua River watershed health analysis using weighted overlay (slope, hydrologic soil group, land use) for non-point pollution identification
- Presented at NEURISA Day 2017, UMass Boston Environmental Colloquium (2018), and Fitchburg State Undergraduate Conference (2018)

PERSONAL PROJECTS & VENTURES

Health & Wellness iOS App (Business) 2026 – Present

- Building and launching an iOS application for the health and wellness industry as a business venture, currently in TestFlight beta
- Full-stack mobile architecture using Supabase (backend/auth), Expo Application Services (EAS) for builds, RevenueCat for subscription management, and multiple third-party APIs for data sourcing
- Implemented cron jobs for background data processing and automated content delivery pipelines

Jarvis (Personal Assistant PWA) 2026

- Built a cross-platform Progressive Web App serving as a daily personal assistant on mobile and desktop with push notifications for task management
- Aggregates news headlines across geospatial, finance, and AI from multiple APIs; includes fitness tracking, career roadmap, wealth management, and FIRE (Financial Independence) tracking modules

Geospatial AI Engineering Roadmap 2026 – In Progress

- Working through a structured 15-week curriculum covering deep learning, RAG, multi-agent systems, LLM fine-tuning, and production MLOps; completed async data pipelines (Python/NumPy/Pandas), time series anomaly detection, and satellite image compression via SVD so far
- Full roadmap: ai-engineer-roadmap-ee841.web.app/projects

EDUCATION

Antioch University New England | Keene, NH

Master of Science, Conservation Biology • Certificate, Applied GIS August 2018 – May 2020

Coursework: Biostatistics, Advanced GIS, Applied Spatial Analysis, Climate Change Resiliency

Fitchburg State University | Fitchburg, MA

Bachelor of Science, Environmental & Earth Sciences • Summa Cum Laude, 3.99 GPA August 2014 – May 2018

Minor: Geographic Sciences & Technology • President's List all semesters

Coursework: GIS, Computer Science, Computer Cartography, Geomorphology